

Jaya Prakash Ravella

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PROFILE

Experienced Software Engineer with 8+ years of expertise in designing, developing, and scaling cloud-native, high-performance web applications and APIs using Java, Spring Boot, Spring Data JPA, Spring MVC, and Micronaut. Skilled in building secure, scalable, mission-critical microservices, implementing RESTful and FHIR-compliant APIs, and optimizing distributed systems for high throughput and low latency. Hands-on experience with AWS cloud services, including EC2, leveraging multi-region deployments to build highly available, fault-tolerant systems. Strong experience with CI/CD pipelines (Jenkins, Git) to ensure reliable, maintainable, and scalable deployments, including cloud-enabled automation to accelerate release cycles. Strong expertise in event-driven architectures using Kafka and RabbitMQ, as well as reactive programming (Project Reactor). Extensive experience with databases and ORM frameworks, including Oracle, PostgreSQL, SQL Server, MongoDB, and Hibernate/JPA, with a focus on writing optimized queries and designing robust data access layers. Proficient in system observability and monitoring using Splunk, Log4J, and custom dashboards to proactively detect and resolve issues. Adept at Agile (Scrum/Kanban) methodologies, TDD, BDD, API standardization, and performance tuning. Proven track record of improving system reliability, reducing defects, accelerating delivery cycles, and delivering enterprise-grade, compliant, high-availability applications.

TECHNOLOGIES & TOOLS

Programming Languages

Java (8/11, Streams, Lambda), SQL, Reactive Programming

Backend & Middleware

Spring Boot, Spring Data JPA, JAX-RS, JAX-WS, Spring Framework (4.x/3.x), Spring AOP, Spring Security, JPA, JDBC, Micronaut

Testing

JUnit, Mockito, PowerMock, Log4J, Postman, Cucumber/JBehave, Test Driven Development (TDD)

Monitoring and Observability

Log4J, Custom Dashboards, Centralized Logging Practices

Version Control & Collaboration

Git, GitHub, SVN, Jira, Agile, Scrum

APIs & Web Services

REST APIs, OAuth1 , OCI

AI/ML

Basics of Machine Learning, Prompt Engineering, OpenAI APIs, LLM Concepts

Pipeline

Jenkins, Maven

Cloud & DevOps

AWS, Docker, Kubernetes (OCI), OCI Cloud

Databases & ORM

Oracle (10g/PL-SQL), PostgreSQL, SQL Server (14.0), MongoDB, Cassandra, Hibernate/JPA

PROFESSIONAL EXPERIENCE

Oracle America Inc, *Software Engineer 2*

11/2022 – Present | Kansas City, Missouri

- Contributed to the design and scaling of a nationwide healthcare messaging platform serving ~30% of U.S. hospitals, delivering secure, real-time, HIPAA-compliant communication across distributed microservices.
- Led backend architecture for next-generation patient messaging, designing scalable REST APIs supporting real-time interactions across hospital networks, improving communication efficiency and reliability.
- Designed and developed referral management APIs using Micronaut and domain-driven design, structuring data as semantic objects to enable AI-driven transformation and advanced analytics.
- Re-architected microservices using Micronaut and reactive programming (Project Reactor), reducing system latency by 25% under peak load and significantly improving throughput.
- Built AI-driven backend services for context-aware automated responses, reducing provider workload and improving response times in clinical workflows.
- Implemented semantic indexing and AI-powered search APIs, enhancing data retrieval accuracy and enabling intelligent insights across large healthcare datasets.
- Collaborated closely with data science teams to productionize ML models, integrating model outputs into real-time systems and reducing manual intervention in decision workflows.
- Consumed real-time AES-triggered events via Kafka to automate downstream processing of status and data changes.
- Developed Kafka consumers using Spring Boot/Micronaut, implementing error handling, retries, and event replay for reliability.
- Configured AES subscriptions and Kafka topics, enabling secure, cross-tenancy event delivery using OCI private endpoints.
- Implemented event-driven workflows to update internal systems and trigger business operations based on AES events.
- Improved system performance by leveraging Micronaut's compile-time dependency injection, eliminating runtime reflection, achieving faster startup times, and lowering memory footprint.
- Engineered lightweight, Dockerized microservices, achieving up to 50% faster startup times compared to traditional Spring-based services.
- Developed Kafka event-driven services, enabling high-throughput asynchronous communication across hospital systems.
- Developed FHIR-compliant APIs, ensuring seamless interoperability with EHR systems and improving data exchange reliability.
- Deployed and managed microservices on OCI Kubernetes Engine (OKE), maintaining 99.95% uptime for mission-critical healthcare applications.
- Implemented centralized logging and observability (Log4J, OCI Logging), significantly improving incident detection and mean time to resolution (MTTR).
- Optimized database performance on Oracle and PostgreSQL, writing high-volume SQL/PL-SQL queries.
- Collaborated in Agile (Scrum) teams, delivering scalable, compliant solutions aligned with strict healthcare regulations.

Stack: Java 8/11, Spring Boot, Micronaut, Project Reactor, REST API, FHIR API, Kafka, RabbitMQ, Docker, Kubernetes (OCI), Oracle SQL, PostgreSQL, PL/SQL, JUnit 5, JBehave, Git, Log4J, Agile/Scrum, AI/ML Integration

Cerner Corporation, Software Engineer 1

05/2020 – 10/2022 | Kansas City, Missouri

- Enhanced the Referral Management Solution, a mission-critical application used by hundreds of healthcare providers nationwide.
- Engineered core backend services using Java 8, JAX-RS, and Jersey 2.x to streamline patient referral workflows, improving care coordination and reducing administrative delays.
- Designed and developed scalable RESTful APIs using JAX-RS and Jersey, integrating with third-party EHR systems and external client platforms.
- Utilized AWS cloud infrastructure for multi-region service deployment, enhancing high availability, fault tolerance, and global scalability of the referral platform.
- Implemented robust data access layers with Hibernate and JPA, optimizing database interactions and ensuring reliable persistence for high-volume healthcare data.
- Standardized API contracts using OpenAPI 2.0 (Swagger), improving client onboarding and integration efficiency.
- Migrated legacy applications to modern Spring Boot architecture, reducing deployment time and improving maintainability.
- Implemented event-driven workflows using RabbitMQ, enabling asynchronous processing of high-volume referral data.
- Developed BDD-style integration tests with JBehave and JUnit 5, achieving 95% code coverage for critical workflows.
- Optimized data transformation with MapStruct and Java Streams, reducing processing time by 25%.
- Developed custom Splunk dashboards for real-time monitoring and proactive defect resolution.
- Utilized CI/CD pipelines with Jenkins and Git for automated builds, testing, and deployments across client environments.

Stack: Java/J2EE, Java MVC, Hibernate, REST API, AngularJS, HTML5, CSS3, JavaScript, AJAX, JSON, Oracle SQL, PL/SQL, Jenkins, Git, Agile/Scrum

Capgemini Technology Services, Software Engineer

08/2016 – 07/2018 | Bengaluru, Karnataka

- Developed and maintained scalable web applications using Java MVC architecture and Oracle Database, ensuring seamless integration, high performance, and reliability in high-traffic environments.
- Designed and built a custom internal enterprise application for Celcom (Singapore) using J2EE, Hibernate, AngularJS, JavaScript, HTML5, CSS3, AJAX, and JSON, delivering a responsive UI and efficient, scalable backend system.
- Integrated RESTful APIs to consume real-time data from external web services, enhancing application functionality and improving overall user experience by 20%.
- Adopted Agile (Scrum) methodologies across the SDLC, collaborating with cross-functional teams and leveraging Jenkins for CI/CD to accelerate development cycles and ensure high-quality, reliable releases.
- Developed and deployed J2EE-based modules on production servers, configuring application servers, managing deployment pipelines, and ensuring smooth release processes with minimal downtime.
- Optimized database and application performance by tuning PL/SQL queries, implementing efficient transaction handling, and maintaining high reliability for enterprise-scale applications. Stack: Java, J2EE, Java MVC, Hibernate, Oracle Database, REST APIs, AngularJS, JavaScript, HTML5, CSS3, AJAX, JSON, Jenkins, Git, Agile/Scrum

Stack : Java, J2EE, Java MVC, Hibernate, Oracle Database, REST APIs, AngularJS, JavaScript, HTML5, CSS3, AJAX, JSON, Jenkins, Git, Agile/Scrum

EDUCATION

University of Missouri - Kansas City,

05/2020

Master of Science (Computer Science)

Relevant Courses: Advanced Software Engineering, Cloud Computing, Principles of Big Data Management

Koneru Lakshmaiah University,

05/2016

Bachelor of Technology (Computer Science & Engineering)

Relevant Courses: Data Structures, Object Oriented Programming Using C++, Internet Programming